
Case Study 07: Calculator and Marriage Eligibility

Code

```
<html>

  <head>

    <title>Calculator & Eligibility of Marriage</title>

    <style>

      body {

        font-family: Arial, sans-serif;

        background-color: rgb(240, 248, 255);

        margin: 20px;

      }

      h1 {

        text-align: center;

        color: rgba(48, 26, 26, 0.792);

      }

      .container {

        width: 50%;

        margin: 0 auto;

        background-color: rgb(215, 215, 215);

        padding: 20px;

        border-radius: 10px;

        box-shadow: 0px 0px 10px rgb(96, 22, 22);

      }
```

```
form {  
    margin-bottom: 20px;  
    padding: 15px;  
    border: 1px solid rgb(204, 204, 204);  
    border-radius: 5px;  
}
```

```
h3 {  
    margin-top: 0;  
    color: rgb(19, 75, 154);  
    border-bottom: 2px solid rgb(0, 123, 255);  
    padding-bottom: 5px;  
}
```

```
input, select {  
    padding: 8px;  
    margin: 5px 0;  
    width: 95%;  
    border: 1px solid rgb(204, 204, 204);  
    border-radius: 4px;  
}
```

```
button {  
    background-color: rgb(40, 167, 148);  
    color: white;  
    padding: 10px 15px;  
    border: none;  
    border-radius: 4px;
```

```
    cursor: pointer;
    font-weight: bold;
    margin-top: 10px;
}
```

```
button:hover {
    background-color: rgb(33, 136, 56);
}
```

```
.calc-btn {
    width: 22%;
    background-color: rgb(0, 13, 255);
}
```

```
.result-box {
    margin-top: 10px;
    font-weight: bold;
    color: rgb(214, 51, 62);
    font-size: 18px;
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<div class="container">
```

```
<form>
```

<h3>Calculator</h3>

<label>Enter 1st Number:</label>

<input type="number" id="num1" placeholder="e.g. 10">

<label>Enter 2nd Number:</label>

<input type="number" id="num2" placeholder="e.g. 5">

<button type="button" class="calc-btn" onclick="calculate('+')">+</button>

<button type="button" class="calc-btn" onclick="calculate('-')">-</button>

<button type="button" class="calc-btn" onclick="calculate('*')">*</button>

<button type="button" class="calc-btn" onclick="calculate('/')">÷</button>

<div class="result-box" id="calcOutput">Result: -</div>

</form>

<form>

<h3>Marriage Eligibility</h3>

<label>Select Gender:</label>

<select id="gender">

<option value="m">Male</option>

<option value="f">Female</option>

</select>

<label>Enter Age:</label>

<input type="number" id="age" placeholder="e.g. 21">

<button type="button" onclick="checkMarriage()">Check Status</button>

```
<div class="result-box" id="marriageOutput">Status: -</div>

</form>

</div>
```

```
<script>
```

```
// Using Logic Arrow Functions
```

```
// 1. Calculator Logic
```

```
let add = (a, b) => a + b;
```

```
let sub = (a, b) => a - b;
```

```
let mul = (a, b) => a * b;
```

```
let div = (a, b) => a / b;
```

```
let calculate = (operator) => {
```

```
    let n1 = Number(document.getElementById("num1").value);
```

```
    let n2 = Number(document.getElementById("num2").value);
```

```
    let res = 0;
```

```
    if(operator === '+') res = add(n1, n2);
```

```
    else if(operator === '-') res = sub(n1, n2);
```

```
    else if(operator === '*') res = mul(n1, n2);
```

```
    else if(operator === '/') res = div(n1, n2);
```

```
document.getElementById("calcOutput").innerHTML = "Result: " + res;  
};
```

```
// 2. Marriage
```

```
let getStatus = (g, a) => {  
  if (g === "m" && a >= 21) {  
    return "Eligible for Marriage";  
  }  
  else if (g === "f" && a >= 18) {  
    return "Eligible for Marriage";  
  }  
  else {  
    return "Not Eligible";  
  }  
};
```

```
let checkMarriage = () => {
```

```
  let gender = document.getElementById("gender").value;  
  let age = Number(document.getElementById("age").value);
```

```
  let resultMessage = getStatus(gender, age);
```

```
document.getElementById("marriageOutput").innerHTML =  
"Status:"+resultMessage;  
};
```

Output

The screenshot displays a web application with two distinct forms on a light blue background. The first form, titled "Calculator", has a header with a blue underline. It contains two input fields for numbers, each with a placeholder "e.g. 10" and "e.g. 5" respectively. Below these are four blue buttons with white symbols for addition (+), subtraction (-), multiplication (x), and division (÷). A red label "Result: -" is positioned below the buttons. The second form, titled "Marriage Eligibility", also has a header with a blue underline. It features a "Select Gender:" dropdown menu currently showing "Male", and an "Enter Age:" input field with a placeholder "e.g. 21". A green "Check Status" button is located below the age field, and a red label "Status: -" is at the bottom of the form.